

Lecture 18: Marginal PMF, Correlation, Covariance, Joint Gaussian Random Variables

1 Marginal PMF

1.1 Definition

For discrete random variables X and Y , the joint PMF is:

$$p_{XY}(x, y) = P(X = x, Y = y)$$

Derivation of Marginal PMFs

The marginal PMFs are obtained by summing over the other variable:

$$p_X(x) = \sum_y p_{XY}(x, y)$$

$$p_Y(y) = \sum_x p_{XY}(x, y)$$

Key Idea (from slide)

The marginal PMF of one variable is obtained by summing the joint PMF over all possible values of the other variable.

1.2 Marginal PMF from a Joint PMF Table

From the joint PMF table:

Row sums $\rightarrow p_X(x)$

Column sums $\rightarrow p_Y(y)$

Formally:

$$p_X(x_i) = \sum_j p_{ij}, \quad p_Y(y_j) = \sum_i p_{ij}$$

Also:

$$\sum_i p_X(x_i) = 1, \quad \sum_j p_Y(y_j) = 1$$

This establishes that marginal PMFs are valid probability distributions.

1.3 Solved Example

Given Joint PMF Table (page 5)

Step-by-Step Solution

Compute $p_X(x)$

$$p_X(0) = \frac{1}{8} + \frac{3}{8} = \frac{1}{4}$$

$$p_X(1) = \frac{3}{8} + \frac{5}{8} = \frac{1}{4}$$

Compute $p_Y(y)$

$$p_Y(0) = \frac{1}{8} + \frac{2}{8} = \frac{3}{8}$$

$$p_Y(1) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$$

Final Marginal PMFs:

$$p_X(x) = \begin{cases} \frac{3}{8}, & x = 0 \\ \frac{5}{8}, & x = 1 \end{cases}$$

$$p_Y(y) = \begin{cases} \frac{1}{2}, & y = 0 \\ \frac{1}{2}, & y = 1 \end{cases}$$

2 Correlation

2.1 Definition

A basic way to measure joint behavior is through the product moment:

$$R_{XY} = E[XY]$$

Interpretation

It gives the average value of the product XY .

It is referred to as a raw correlation measure.

2.2 Limitations of R_{XY}

From the slides:

- Depends on the scale of X and Y
- Computed about the origin, not about the means

Therefore, it mixes:

- Mean values
- Spread
- Dependence

Important Observation

$$R_{XY} = 0 \Rightarrow \text{orthogonal about the origin}$$

2.3 Transition to Covariance

To remove the effect of means:

- Center the variables first

This leads to the concept of covariance.

3 Covariance

3.1 Definition and Interpretation

Covariance is defined as:

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

Reasoning Behind Definition

1. Subtract the mean from each variable
2. Measure how the centered quantities vary together

Interpretation (from slides)

Covariance captures whether deviations:

- Occur in the same direction $\rightarrow$ Positive
- Occur in opposite directions $\rightarrow$ Negative
- Have no linear relation $\rightarrow$ Zero

It indicates the direction of joint linear variation.
However:

- Its magnitude depends on units
- It cannot measure strength of relationship

3.2 Algebraic Form and Derivation

Start with:

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

Expand the expression:

$$= E[XY - X\mu_Y - Y\mu_X + \mu_X\mu_Y]$$

Substitute:

$$\mu_X = E[X], \quad \mu_Y = E[Y]$$

$$\text{Cov}(X, Y) = E[XY] - \mu_Y E[X] - \mu_X E[Y] + \mu_X \mu_Y$$

$$\text{Cov}(X, Y) = E[XY] - \mu_X \mu_Y$$

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

3.3 Properties of Covariance

1. Symmetry: $\text{Cov}(X, Y) = \text{Cov}(Y, X)$
2. Variance as Special Case: $\text{Cov}(X, X) = \text{Var}(X)$
3. Linearity with Constants: $\text{Cov}(aX + b, cY + d) = ac \cdot \text{Cov}(X, Y)$
4. Independence Implies Zero Covariance: $X \perp Y \Rightarrow \text{Cov}(X, Y) = 0$

Important Note

Zero covariance means uncorrelated, but does not imply independence.

3.4 Example 1

Problem

If:

$$Y = -2X + 3$$

Find $\text{Cov}(X, Y)$.

Step-by-Step Solution

Using:

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

Step 1: Compute XY

$$XY = X(-2X + 3) = -2X^2 + 3X$$

Step 2: Compute expectation of Y

$$E[Y] = E[-2X + 3] = -2E[X] + 3$$

Step 3: Substitute into covariance formula

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

4 Pearson's Correlation Coefficient

Definition

Measures the effect of change in one variable when the other changes.

Properties

- Measures strength and direction of linear relationship
- Normalized version of covariance
- Unit-free

5 Jointly Gaussian Random Variables

5.1 Definition

A pair (X, Y) is jointly Gaussian if:

- The joint density has the bivariate Gaussian form
- The marginals are Gaussian

5.2 Applications (from slides)

- Signal processing
- Communication systems
- Modeling continuous noisy measurements

5.3 Example (from slide)

Scenario

X : Atmospheric CO_2 concentration

Y : Global surface temperature anomaly

Because both vary due to many random processes, their joint behavior can often be approximated by a joint Gaussian distribution.

5.4 Algebraic Form and Interpretation

(Referenced in slides as continuation; algebraic form and interpretation follow from the definition of joint Gaussian density.)

6 Final Logical Flow Summary

1. Start with joint PMF
2. Derive marginal PMFs via summation
3. Measure joint behavior using $E[XY]$
4. Identify limitation $\rightarrow$ depends on means
5. Introduce covariance by centering variables
6. Derive algebraic form and properties
7. Normalize $\rightarrow$ Pearson's coefficient
8. Extend to continuous case $\rightarrow$ Joint Gaussian RVs